

1 – Discussion

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1.1 Description of the Problem



Halliwell befriending service is a charity which provides various services to the elderly in the Halliwell area. They focus on reducing isolation in the community by running activities which include: luncheons, games, activity sessions, a computer club, and a craft club. They also visit the elderly in their homes for two hour visits.

It first began when Fiona and Joan saw a large community of increasingly isolated over 50s in their area and wanted to do something about it. They saw these people getting little help from the community. They managed to get some funding from the City Challenge Funding and began a small befriending service. This was in 1994 and since then has grown from just two volunteers to over 80 and now operates across Bolton in Halliwell, Astley Bridge, Heaton, Smithills and Sharples. As the needs of the elderly have changed, the befriending service has adapted to meet these needs. They're currently funded by Bolton Council Adult

Services, Bolton at Home, Bolton CVS and Ambition for Ageing.

Originally, it was based at the Free Church on Halliwell Road but, since 2014, it has been based at St. Paul's Community Room, which is also on Halliwell road.

Most of the activities run fortnightly. On Monday, there is a luncheon club and a games and activities afternoon. There are also various coffee mornings. Home visits are Monday to Friday between 0900 and 1600. There is also a drop-in session on the third Thursday of each month, from January to November.

I'm going to be focusing on the computer club and this runs every Thursday morning. The computer club has the aim of helping the elderly use computers so that they are able to use online banking and apply for various passes they may need e.g. the blue badge. This is done using an online system, which provides multiple 'lessons' for the users to complete. Each lesson walks the user through a different type of form entry. There are lessons for password fields, radio buttons, and dropdowns, for example. Each lesson shows a section of a form. All of the fields, other than the one for the user to practise, are already filled in.

The main problem is that the current system doesn't provide enough practice of filling forms in from scratch. As mentioned before, the forms have a large amount of data already filled in so the user is not able to practise with a full form. This means that, when the users have to actually fill in a form from scratch, they're not properly prepared as they haven't entirely done it before.

The forms provided don't provide more complex interactions for the users to practise, e.g. uploading an image or attaching a document and these are vital skills that they need to practise as they are a key part of a large amount of online forms, especially those which are used to apply for things such as railcards. The user doesn't have the full experience such as logging into a site, finding the relevant form, filling it in from scratch, and submitting it. Because of this, they



don't feel confident doing it on their own. These factors mean that, even though the user has practised certain question types before, they may still struggle to fill in the form.

The system provides no means of sorting through practice questions nor is there a database of practice questions. This means that users can't find specific questions. Currently, if they want to practise a particular type of question, they have to manually search through each lesson to find it. This is tedious and time consuming. This organisational problem means that users are often left without enough practice on the questions that they currently struggle with and impedes their ability to fill in forms themselves. The system also is unable to generate practice forms so the user is limited to the few hard-coded forms that are in the system and are left with little practice once they have exhausted these.

The current system also doesn't track progress in a more meaningful way than just the number of forms completed. It doesn't time how long it took each user to complete the form. It also doesn't break down the questions completed by type or highlight the types of questions struggled with. Because no metrics other than number of forms completed are tracked, the current system is unable to generate forms tailored to each user's weaknesses. It also means that the volunteer running the session can't check to see how each user is doing without asking them. This creates a problem as the user may not remember what they struggled with or may not be able to accurately describe it.

The current system is audio based. The system speaks the instructions to the user on how to complete the form. As there are many users all in a small room, the noise level is quite high and this makes it difficult to understand what the system is saying because there are so many speaking at the same time. The problem is compounded because the users are elderly and they have difficulty hearing to begin with so it makes it even harder to understand what the system is asking them to do. This communication problem means that sometimes instructions have to be repeatedly played in order to be understood and slows the pace of the sessions as time is wasted trying to understand what the system is saying.

I am going to gain and use feedback in a few ways. Firstly, I am going to gather feedback from my contact (Steph Stather) as she is the volunteer who runs the computer group therefore she knows exactly what the new system should be able to do and will be able to give me very relevant feedback. As she will be the one that is actually using the system in practice, she will be able to tell me exactly how the system should be tailored to her particular use case. This will allow the system to be as bespoke as possible and fully satisfy the requirements. I'll gather this information through phone calls, emails, and meeting face to face.

I am also going to gather some feedback from my peers in my class. This is because they are knowledgeable in the technical elements of computer programs and software, which means that they will be able to give me in-depth and technical feedback on the system that I am developing and will be able to suggest changes that could possibly make it better. I am going to present the aims (that I am going to detail later in the document) to them to gather their feedback.

Contacts

- Steph Stather

Steph Stather is a volunteer at the befriending service. She leads the computer group, helps at the craft group, and is a secretary on the management committee.

She volunteers regularly at the befriending service and is a key part of it.

She volunteers every Thursday morning to run the computer club with other volunteers. She helps out at other groups on a Tuesday afternoon but she isn't the leader of these groups. She also attends management committee meetings and these run twice a month.

1.2 Broad Aims of the Planned Solution

Aims:

- Ability to search through and create practice questions and create forms based on them.
- Ability to identify questions that the user struggles on.
- The forms should have realistic and rigorous validation.
- The system should have an easy to use user interface.

Ability to search through and create practice questions and create forms based on them

This is an important feature of the system. There should be a database of all the practice questions in the system. Each question should have information about its type and difficulty, for example, stored about it. A type of sort such as a bubble sort would be used to sort based on any of these fields. I would also implement a search algorithm, such as a linear search, to allow the user to search based on key terms. Once they have found the question that they want to practice, the user should be able to press a button to open a new window, in which the question would have been instantiated, allowing them to practice it right away.

If a user wants to practice more than one question, the system should allow them to select multiple questions and, from these questions, should generate a custom form for them to fill in. As the user selects questions, they should be added in turn to a blank window, and thus create the form. The system should take care of all of the formatting and layout of the form so all the user has to do is select the question types that they want. This will help the user practice filling forms in because they will have a large bank of practice questions and will be able to create forms that are tailored to them.

The admin user should be able to create questions to be added to the questions database and so can be added to forms. This will extend the life of the system as not all of the questions have to be made by the developer. The questions should be able to be exported to a file and then imported by other instances of the system on other computers. The exported question would first have to be copied onto the other computer.

The admin should be able to create these questions by an easy drag and drop interface. This will enable them to create new questions quickly and easily.

This addresses one of the main issues with the current system because it provides a large amount of easily accessible practice questions and forms for the user.

Ability to identify questions that the user struggles on

The system should be able to time how long it takes a user to complete each question and how many times they failed the validation check. This will provide useful data on which questions the user is struggling with and which questions they're proficient with. This type of data is invaluable to helping them progress

because it quickly highlights the areas that the user needs to improve on. The aim of the computer club is to help the elderly become better at using computers. Being able to find each user's weak areas expedites this process because they can easily see where they need to improve. It would also allow the system to generate forms based on each user's weaknesses.

This data should be permanently stored in a text file and updated / appended to whenever a user completes a form.

The system should produce a report detailing the types of questions most commonly failed and statistics on how long it takes them to complete a form. These could be, for example, the time it takes them to complete each type of question and average number of times that they fail the validation checks. The data from the text file should be loaded into order to produce this report.

A user should only be able to create a personal report, not a report that contains information about other users. The report should detail the questions that they have been struggling on based on the data stored. The admin user (the volunteer) should be able to log into the system and request a report detailing the weaknesses of all the users in general. They should also be able to produce and view a report for each individual user.

The forms should have realistic and rigorous validation

Validation is an integral part of online forms therefore, in order for my system to provide the users with realistic forms to practise, my system must also include validation. The system should include types of validation such as format checks, length checks, lookup checks, presence checks, and range checks. These types, of validation, would be used for the validation of email-addresses, phone numbers, and addresses, for example. There should also be types of verification such as double entry verification on passwords. These are all in the aim to make the forms as realistic as possible to give the users the best experience. Without validation, the users wouldn't be able to practice very well because, even if they had made an error in the form, it wouldn't be highlighted to them, and this means that they could have filled in the entire form wrong without knowing and this doesn't help them to improve.

The system should have an easy to use user interface

The system's primary purpose is to help the elderly learn how to use computers. When they start using the system, they will have very few computer skills and will find computers difficult and confusing to use. Because my system is targeting those users that aren't able to use a computer, it's essential that the system is well laid out, clear, and easy to use, otherwise, these inexperienced users will find it incredibly difficult to use. Furthermore, if the system isn't easy to use, the volunteer will have to waste time helping each user operate the system and this takes time away from helping users to fill in forms - the actual aim of the sessions.

The system should have a simple interface. This is to avoid overwhelming the users and potentially making them have very little confidence in using computers as this will slow their progress. Large amounts of nested menus, panels, and windows will make the system unnecessarily complex and will make it difficult to use. The system should only have a few windows and few menus. There should be limited nested options and all of the actions that the user may want to take (for

example generating a form) should be clearly presented to them with large and clearly labelled buttons and menus. This is to make the system as approachable and as easy to use as possible.

There should be a context sensitive help screen provided in each screen of the system. There should be a help button available in every screen of the system which the user can press which brings up a window that tells them how to use the part of the system that is currently open. For example, if they pressed the help button from the search question window, it would explain to them how to search for questions.

The system could potentially have Halliwell befriending service branding however this isn't entirely necessary and wouldn't add much, if anything, to the system. Furthermore, Halliwell befriending service doesn't have much of a brand, for example they don't have a logo, therefore there wouldn't be much branding that could be added to the system.

1.3 Limitations of the Planned Solution

- I think that it would be helpful if the system was able to communicate with other computers running the system over a LAN (Local Area Network). This way, the volunteer could run the system on their computer and monitor how each user is performing. They could also assign tasks for each user to do. If a user is struggling, this could be highlighted to them. I won't be able to do this for a number of reasons, however. Firstly, the programming required would be complicated and a little beyond my current skill level because I am not competent using sockets, for example, and networking also introduces a whole host of new software bugs. This doesn't mean that I wouldn't be able to do it but it means that it would take me longer to develop the system and this extra time most likely wouldn't fit in the time constraints.
- A feature that I think would help the users considerably would be if the system could parse a webpage and extract the form elements to recreate the form for practice. This means that the system could be given the HTML of a form on a website and it could recreate the form so that it could be practised over and over. This would mean that the users could easily practice realistic forms that were almost identical to the real ones with little work on their part. However, I don't think that this would be possible because there are many different ways that forms are implemented, e.g. pure HTML or HTML and JavaScript, on websites. This means that it would be time consuming and difficult to make a system that would be able to accurately parse the forms. I'm also not very familiar with HTML forms therefore there would have to be a large amount of additional learning that I would have to do which could delay the project and make it not fit within the time constraints.
- A mobile app version of the system would be good as it would allow the users to learn how to fill in forms on a smartphone. This is increasingly relevant as mobile phones are being used more and more as people's primary computers and they are more than capable of simple form filling tasks. Many elderly computer users struggle to use smartphones and are more used to older, less advanced, phones. A mobile app version of the system would address this area of skills that they lack and would mean they would be able to use their phones to their fullest. However, I won't be able to make a mobile version of the system because I am unfamiliar with making Android and IOS apps. Furthermore, I don't own an iPhone to test the IOS app on, a mac to develop the app, or the expensive developer license required to publish the app in the app store. Producing an Android app may be easier as I have all the tools needed however I would be able to reuse very little code from the desktop application and so I would spend a large amount of time rewriting features for mobile. The added cost of the development tools and the added development time means that this feature won't be feasible.
- There are multiple computers at the computer club therefore it would be useful if there was a single sign on system so that each user could sign in to any computer and have their user account synced between computers. This means that, if a user started completing a form on one computer, the user could complete the form on another computer and have their progress tracked between the computers. However, I think that this wouldn't be feasible because of the same reasons mentioned above about my inexperience with networking. In this case, there would need to be a server that runs the system and stores the user details. There is no computer at the computer club that would be able to be used as a central server therefore one would have to be purchased and this isn't feasible because it would be a large expense for the charity and there are better ways that they could use the

money. There is also no one that is experienced enough to be able to maintain the server. I'm also not experienced in writing server applications, only client-side ones. Therefore, I am unable to implement this feature because it would be too costly and too complicated to program and maintain.

- I could produce a series of walkthrough videos that go with the system and walk the users through how to complete each form. This would be useful as it would mean that the users could independently learn without the volunteer having to help them through each form. It could also mean that the more able users would be able to work through harder elements on their own without needing to wait for the rest of the group to catch up. I don't think that I will be able to implement this into my system because I don't have the facility necessary to produce high quality tutorial videos. A good microphone and an isolated recording area would be needed in order to ensure that my voice is as clear as possible so that they have the best chance of being able to understand what I am saying. This is important as a problem with the current system is that it's difficult to understand the audio walkthrough of each form as the environment can become noisy. I'm also unable to do this because creating these walkthroughs would take a large amount of time and so would mean that I wouldn't meet the time constraints.
- It would be useful if the system recreated entire websites. This would mean that the user had the full experience of logging into the website and locating and filling in the form which allows them to develop these essential skills. The system as realistic as possible because it would include almost carbon-copies of the most common websites that they'd need to fill forms in on e.g. applying for a disability badge. I won't be able to implement this feature because I am not competent enough in developing graphical user interfaces to be able to recreate complicated websites, like these, to a sufficient standard. Also, many elements of a website's GUI are easy to create in HTML and CSS but not in desktop programming languages for example animated dropdown lists. Furthermore, I would have to spend a large amount of time locating the forms that the user would need on the website and recreating them. I could either recreate only the screens/pages of the website that the user would see when they were looking for the form or I could recreate the entire website. The former would mean that the user had no ability to 'get lost' on the website as only the pages that they would see if they were navigating it correctly would be programmed and this means that the user has no opportunity to get it wrong therefore they may not learn very much. The latter would take a large amount of time as some websites are incredibly large and complicated. Recreating each page would simply take too long and be too tedious.

1.4 Consideration of Feedback

Feedback	From	Accept/Reject	Action to be taken (including Justification)
"Find the average time that is take for each question to give a more accurate time"	Ben Canning (peer)	Accept	I will add to the time recording feature so that stores the previous times that the user has taken to complete the question. This will allow the system to calculate the average time that it takes for the user to complete the question. This will be a useful feature of the system because the mean of the times will be more accurate and more representative of how long it takes them to complete a question compared to a single time in isolation. This allows the users to better see how they are progressing because they can see if the time taken for them to complete a question was above or below the average time that it takes them and it means that they can easily see if they're improving because, if they are, their average time will decrease.
"have the basics on one tab which sends the user to another tab with the corresponding what they want to access"	Ben Canning (peer)	Reject	This will add a large amount of nested menus. Rather than having buttons on one tab which then send the user to another tab to perform the action that they want to access, I think that it would be more appropriate just to allow them to perform the actions from the original tab. This would make the interface less complicated and easier to use.
"You could create a scoring system to add a sense of competition and give them motivation to improve"	Sam Ball (peer)	Reject	I think that this is a feature that wouldn't be very useful for the system. This is because the users are at the club to improve themselves and it might take each user quite a while to become familiar with a computer therefore they might not be comfortable competing with the other users. This might be stressful for them. In addition, it would be difficult to make a central scoreboard system as the computers would have to be networked and this is beyond my current skill level.

“Offer rewards for completing forms e.g. vouchers for online shopping to incentivise use of online forms”	Sam Ball (peer)	Reject	While this could give the users a good reason to want to improve, there is no facility for me to get vouchers like this and, if the befriending service were to buy prizes, it would be a large and mostly unnecessary cost. Because the befriending service is a charity, money is limited and needs to be spent well.
“Graph the results of average scores to compare previous attempts to the current attempt, as to be able to see progress.”	Matthew Hodgson (peer)	Accept	This would be a good feature of the system and would allow the users to easily see their progress. However, it would be difficult to implement the graphing functionality if I programmed the graphing functionality from scratch. This is because, it would take a long time and would probably be above my skill level. I could use a premade graphing library and integrate it with my system. This would be simpler and allow me to much more easily add the functionality to the system.
“A UI that adapts with the user, introducing new elements as they score better scores - could be linked with the graph, marking new unlocks on the graph, giving the users clear aims to work towards.”	Matthew Hodgson (peer)	Reject	This would be a good feature because it would allow the UI to adapt with the user. The more complicated functions could be hidden when the user first starts using the program. This would make the software as simple as possible as not all of the functionality would be available. As the user gets better, they could use more complicated features. However, I won't be accepting this feedback because the UI changing regularly could be very confusing for the users. It could also overwhelm them if, just when they got used to the UI, it changed.

"I'd recommend having an admin/worker login that gives them a bit more access to the backend. Creating new forms or elements of forms, viewing data on what forms/fields are commonly filled in incorrectly (this may allow them to target training courses), etc. This saves it from being too static and allows your system to grow without the intervention of the developer."	Nick Wilkie (teacher)	Accept	A feature like this would be very useful. It means that, as Nick mentioned, the system would be able to have some new handmade forms without the developer needing to edit the system. Because the user will already be able to create their own forms (to some extent), I think it would be most useful for the admin (volunteer) to be able to create a form on their computer running the system and then export it so that it can be imported by all the other computers that are running the system. This means that the users wouldn't have to create the form themselves.
"Include a register system so that you can easily track user attendance and map against user attainment"	Nick Wilkie (teacher)	Accept	A register system would be a useful addition to the system. This is because it would allow more detailed progress tracking. Mapping user attendance to user attainment would allow weeks where the user wasn't present to be marked on the main progress tracking graph to see how absences have affected progress. To have a good register system, other personal details would need to be stored, such as phone number and date of birth. The first names and surnames of all the users would also have to be stored. This extra information enables the progress of the user to be put more in context and so a clearer picture of how they're doing can be seen. For example an older user may not progress as fast as a younger user.



These images show me presenting my ideas to my peers and gathering feedback. I made a presentation which can be seen on my computer's screen. It gave my peers a little background information about Halliwell befriending service, the problem, and detailed the broad aims and limitations that I had come up with so that they could give me feedback about them.

1.5 Revised Broad Aims

Aims:

- Ability to search through and create practice questions and create forms based on them.
- Ability to identify questions that the user struggles on.
- The forms should have realistic and rigorous validation.
- The system should have an easy to use user interface.
- The system should store the personal details of each user

There are few changes to the broad aims of the solution and an entirely new broad aim as a result of the feedback.

Ability to search through and create practice questions and create forms based on them

The system should allow the admin to log into the system and create forms in a similar way to the way in which users create them. The admin should then be able to add this premade form to the database of forms that the user can access. This means that the admin will easily be able to add new forms into the system. If the admin has created a new form on their own computer, or a different computer to that which the user is using, they should be able to export this form and save it onto a USB stick, for example. They should be able to connect this USB stick to another computer that is running the system and, from this, should be able to import the form onto the computer. This would mean that the volunteer would be able to quickly distribute a new form that they have made to all of the computers around the room.

Ability to identify questions that the user struggles on

Rather than only making comparisons based on the most recent time, that it took the user to complete a question, the system should use the stored times from the previous attempts to calculate an average so that the time taken is more accurate. I am going to use a rolling average, for example an average of the past ten attempts, because, this way, the average will be less influenced by earlier scores and will better reflect their current ability.

I'm going to implement some graphing functionality so that the recorded information about how the user is progressing (for example, time to complete questions and number of times that they've failed validation checks) is easier to understand and easier to compare to their past attempts. I could use a line chart to plot the past times that it has taken the users to complete each question. I could also use a line chart to plot how many times the user failed the validation check for each separate attempt of the question over time.

I could use a pie chart to represent what percentage of the different question types that the system provides have been attempted. A pie chart could also represent how many times they fail a validation check compared to how many times they successfully complete a question.

By using the other data that is stored about the users, the system should be able to compare how the user is progressing against more contextual factors. For example, it would be useful to mark the times when the user was absent on the progress tracking graph so that areas of less progress/no progress can easily be identified as session that the user was absent. Storing date of birth means that the user's progress can be compared against their age as an older user may progress slower than a younger user.

This progress report should also be able to be printed. The user's first name and last name should be added the top of the report so that when it's printed it's easy to see which report corresponds to which user.

The system should store the personal details of each user

Details such as the first name and surname of each user should be stored along with their attendance, phone number, and date of birth. This means that the admin will be able to search through the users and easily find information about them. For example, they would easily be able to see who was present at a particular session.

The system should also allow easy attendance tracking. When Steph needs to take the register a list of all the users should be displayed and she should just be able to mark the ones that are present with a check mark.

This data about all of the users should be permanently stored in a text file. This information is sensitive therefore it should be encrypted to prevent access by people who are unauthorised. Only Steph's account (the admin account) should be able to read and decrypt this file. This file would be stored on Steph's copy of the system on her computer.

The attendance of a particular user along with their name and age should also be stored locally on the computer being used by that user. This data should also be encrypted. Only that user should be able to decrypt it. This data would be used when creating the progress reports as mentioned above. This means that data about the user can be accessed by the user without having to decrypt the main database that is on Steph's computer and without having to make a copy of that database to be stored on the user's computer. This system of multiple databases means that there will be data redundancy. However, I think that the increased security of allowing users to access to only the data they need outweighs the disadvantages of data redundancy.